PIC Processor Architecture

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Why do we talk about PIC?

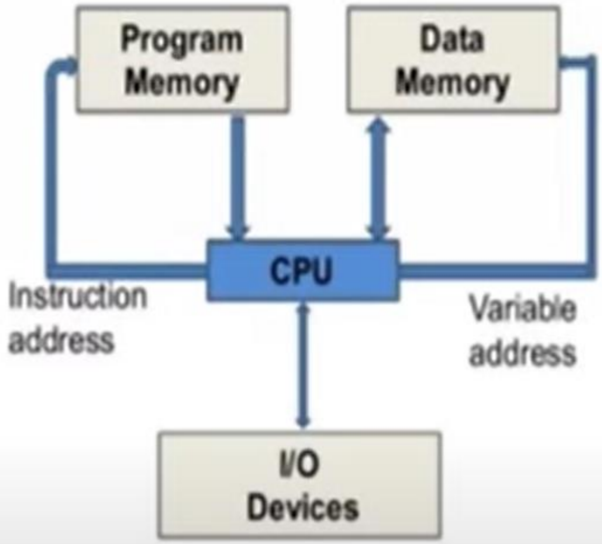
- PIC stands for Peripheral Interface Controller.
- PIC microcontroller was developed by microchip technology in 1993.
- It was developed for supporting PDP computers to control its peripheral devices and that's why it was named Peripheral Interface Controller.
- PIC microcontrollers are of low cost,
- very fast and easy for the programming and execution of program
- Their interfacing with other peripherals is also very easy.

Why do we talk about PIC?

- PIC Microcontroller architecture is based on Harvard architecture
- supports RISC architecture (Reduced Instruction Set Computer).
- PIC microcontroller architecture consists of memory organization (ram, rom, stack), CPU, timers, counter, ADC, DAC, serial communication, CCP module and I/O ports.
- PIC microcontroller also supports the protocols like CAN, SPI, UART for interfacing with other peripherals.

- **Peripheral Interface Controllers (PIC)** is a family Of microcontrollers by *Microchip Technology*.
- They have attractive features and they are suitable for a wide range of applications.
- They are **RISC processors** and use **Harvard architecture**. Harvard architecture makes use of separate program and data memories.
- This has two different effects:

1. The data and address buses are separate allowing increased data flow from the CPU.
2. There can be different widths of these buses. The data memory Of PIC is **8-bit wide**, Whereas the PIC program memory is **12-, 14- or 16-bit wide**.



Harvard Machine

- There are only **35 PIC instructions**, which is easier for the programmer to remember by practice.
- It executes most of its instructions within 0.2 μs, when operated at its maximum Clock rate.
- An interesting thing about PIC is that its machine cycle Consists of **4 clock pulses in contrast with the 12 clock pulses** per machine cycle in Intel 8051.
- PIC instruction set is **highly orthogonal**. The term orthogonal means non-overlapping or mutually independent.
- When an instruction set is well designed and highly orthogonal. fewer instructions are required to perform the required tasks. With fewer instructions, the whole set can be learned easily and quickly, making the programmer's job simple and efficient as well.

- PIC devices include **analog-to-digital converter (ADC)** function, built-in Power-on-reset and Brown-out-reset features.
- **Brown-out-reset** means that when the power supply drops below a certain voltage (4 V in case of PIC), it causes PIC to reset.
- The oscillator frequency determining components of PIC can be a low-cost RC circuit or a quartz crystal or a ceramic resonator. The oscillator options are EPROM selectable.
- PIC supports a **power saving SLEEP mode**. The clock may be frozen with all the data preserved in the processor memory.

ARM architecture

- PIC Microcontrollers from Microchip Company are divided into 4 large families

•First family: PIC10 (10FXXX) called Low End

•Second family: PIC12 (PIC12FXXX) called Mid-Range

•Third family: PIC16 (16FXXX)

•Fourth family: PIC 17/18 (18FXXX)

ARM architecture

- PIC10 and PIC12.
- PIC16.
- PIC24 and dsPIC.
- PIC32M MIPS-based line.

PIC NAMING VERSION

- In family names, there are suffixes used in between which tell -
- PIC18FXXX – ‘F’ implies Flash Program Memory
- PIC16CXXX – ‘C’ implies EPROM Program Memory
- PIC18LFXXX – ‘LF” implies Low Voltage operation
    - LC -UV EPROM Extended
    - CR- Masked ROM standard
    - LCR- Masked ROM Extended
- It varies from family to family.
- Whenever a new part is designed to replace an older part, a new version of the part is obtained by adding extra alphabet at the end like 16F877 and 16F877

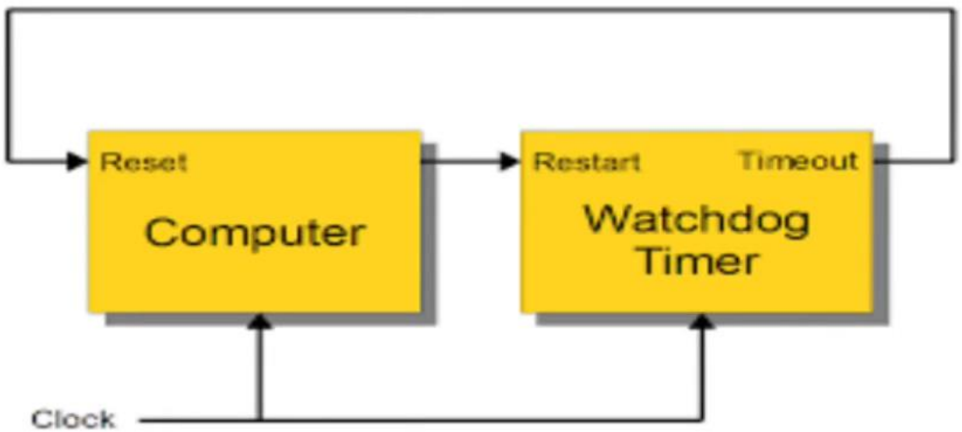
Features of PIC architecture

- 8 Bit microcontroller
- Operating speed up to 20MHz
- Flash program memory 14 bit words, 8KB
- Data memory Data SRAM 368nbytes and data EEPROM 256 bytes
- Power saving sleep mode
- 5 input /ouTput ports
- 15 interrupt signal
- 3 timers
- 35 instructions
- 10 bit A/D module
- CCP module Capture/Compare/PWM (CCP)
- In circuit serial programming via 2 pins
- Watch dog timer with on chip RC oscillator
- **Working register – W register**
- **FSR –File select register**
- **Burn out register**

PIC 16F8XX

PIC MICROCONTROLLER	Program memory	Data memory	No.of instructions	No.of timers	I/os	Other features
16f877 DC to 40 MHz	8K x 14 words	368Bytes of RAM and 256 bytes of EEPROM data memory	35	3+WDT	Ports A, B,C,D, E	8 channel 10 bit ADC Brown out Reset BOR 2 CCP models Power on Reset POR 2 Serial ports 15 interrupts

- PIC has a **Watchdog timer** with on-chip RC oscillator. A watchdog timer *prevents the processor from endless loop hanging condition*.
- Watchdog timer is a **simple timer circuit**, which looks after the continuous functioning of the system with respect to time.
- The processor periodically resets the watchdog timer. This also indicates that the processor functioning is going on.
- However, if the watchdog timer is not reset before it expires, it is a faulty condition and the watchdog timer now resets the processor. Thus it increases the system reliability.
- Maximum 12 independent interrupt Sources are supported by PIC. The operating voltage range typically for 16C7X is 3.0 – 6.0 V. Power consumption is very low.



'able 9.1 PIC parts 16C61, 16C71, 16C66, 16C74

PIC device	Program memory	No. of instructions	Data memory	Timers	I/O	Other/additional features
16C61 Max. speed 20 MHz	1024 × 14 words EPROM	14-bit, 35 instructions	36 bytes User RAM	1 + WDT	13 I/O pins with 25 mA source/20 mA sink per I/O	3 interrupt sources
16C71 Max. speed 20 MHz	1024 × 14 words EPROM	35 instructions	36 bytes User RAM	1 + WDT	13 I/O pins with 25 mA source/20 mA sink per I/O	4 interrupt sources 4 channels of 8-bit ADC
16C66 Max. speed 20 MHz	8192 × 14 words EPROM	14-bit, 35 instructions	368 bytes User RAM	3 + WDT	22 I/O pins with 25 mA source/sink per I/O	Brown-out-reset Power-on-reset 10 interrupt sources 2 capture/compare/ WM modules. 2 serial ports. USART
16C74 Max. speed 20 MHz	4096 × 14 words EPROM	14-bit, 35 instructions	192 bytes User RAM	3 + WDT	33 I/O pins with 25 mA source/sink per I/O	8-channel 8-bit analog-to-digital converter. Brown-out-reset (BOR), power-on-reset (POR) 2 Capture/Compare/ PWM modules 2 serial ports and a parallel slave port. USART

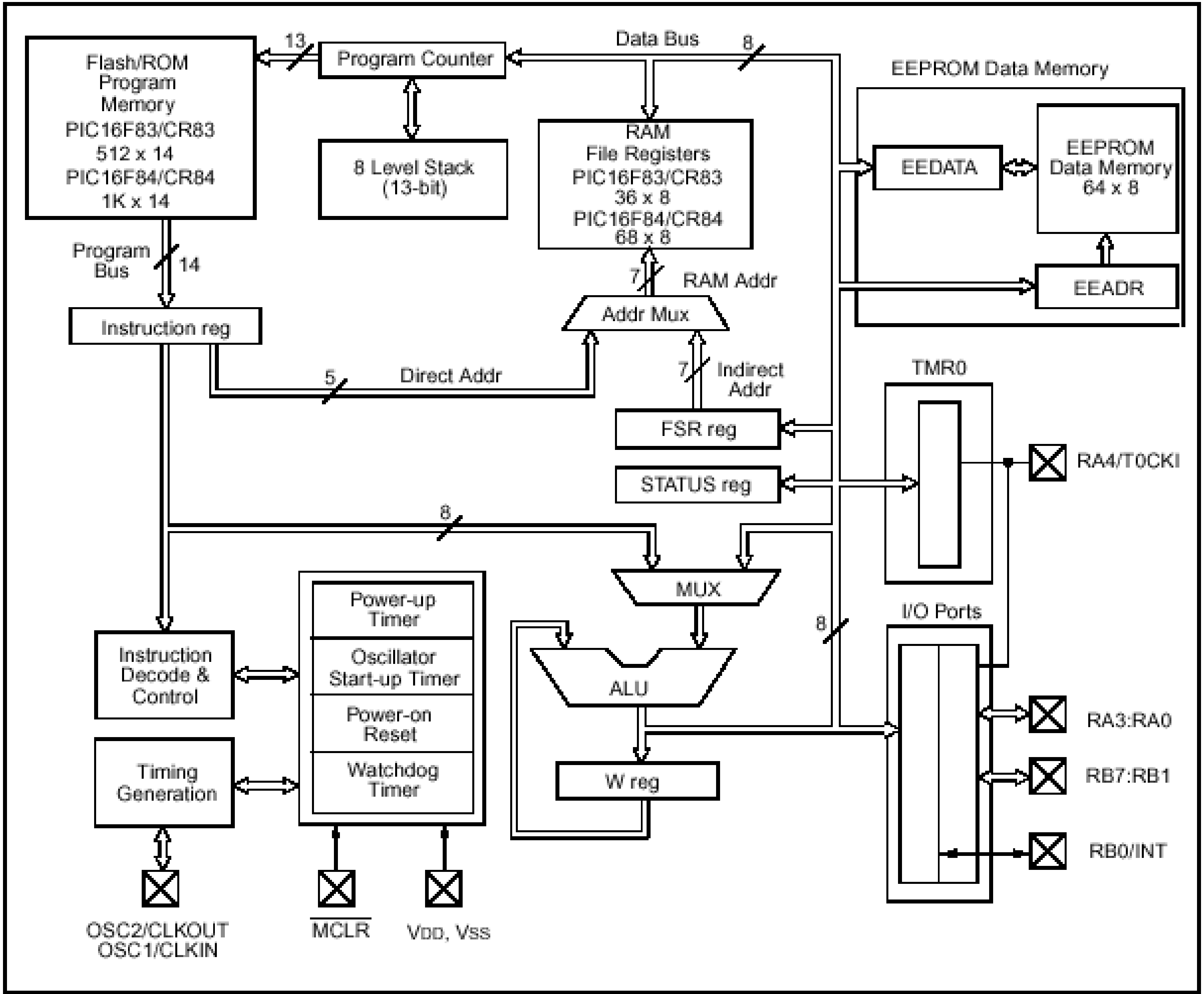
Selection Criteria for PIC Microcontroller

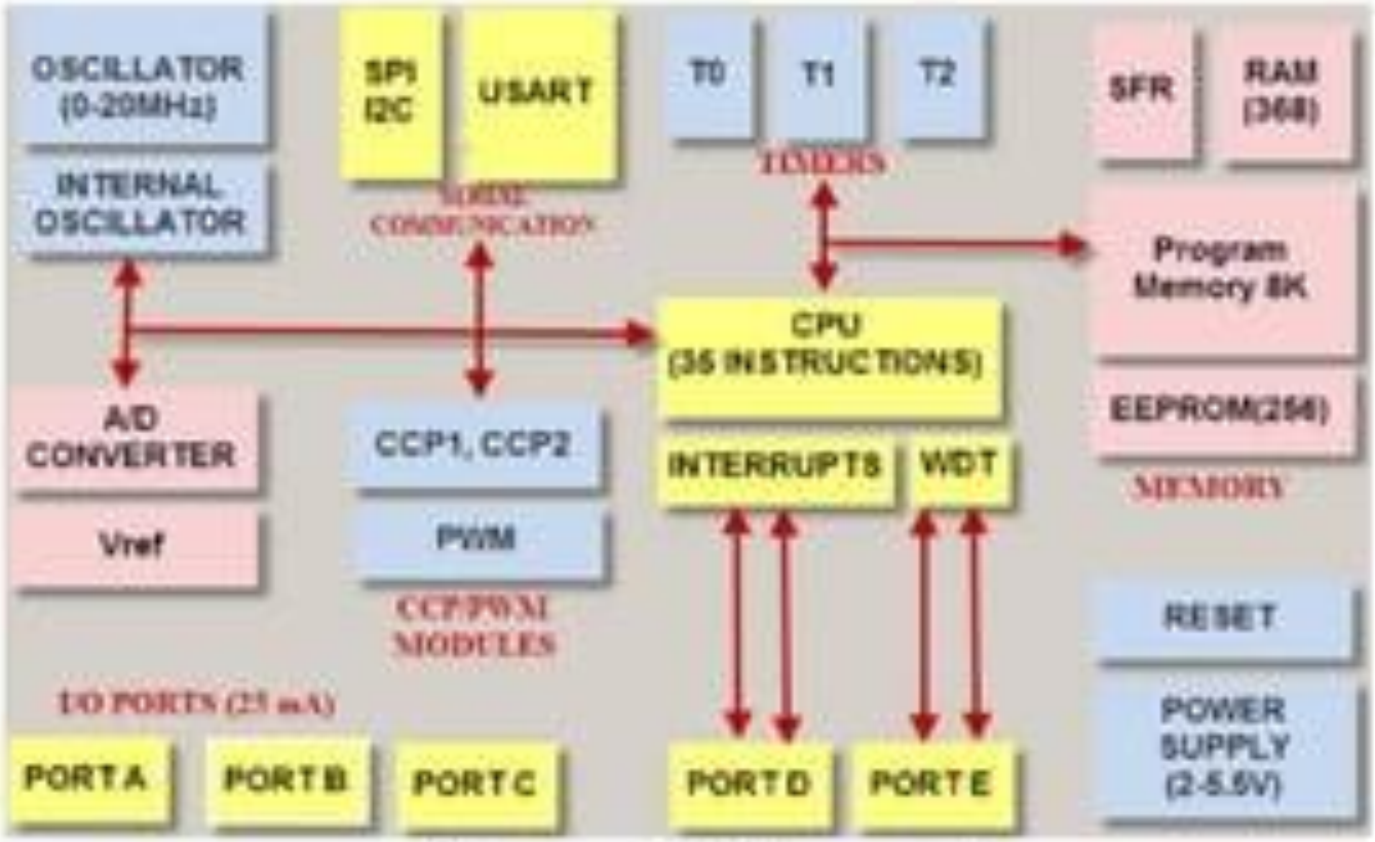
- 1) Speed
- 2) Amount of RAM/ ROM
- 3) Number of I/O pins, Timers
- 4) Power consumption
- 5) Availability of tools
- 6) Added features like ADC/ DAC/ CCP, Bus support like CAN, SPI, I2C, USB.
- 7) Watchdog timer, Timer modes, Data EEPROM etc.

PIC 16F877 – Overall features

- Expanded as Peripheral Interface Controller (PIC)
- Manufactured by **Microchip**.
- Very efficient and at the same time, economical.
- 8 Bit microcontroller.
- Package options (40 or 44 pins)
- RISC architecture.
- 35 Instructions.
- Support for
  - **8K x 14 words of FLASH Program Memory,**
  - **368 x 8 bytes of Data Memory (RAM)**
  - **256 x 8 bytes of EEPROM Data Memory**
- Support for interrupts.
- Direct, indirect and relative addressing modes – Suffice.
- Power saving features.
- Watch dog timer.
- Low power consumption.

FIGURE 3-1: PIC16F8X BLOCK DIAGRAM





Architecture of PIC Microcontroller

CPU (Central Processing Unit):

- PIC microcontroller’s CPU consists of
- Arithmetic logic unit (ALU)
- Memory unit (MU)
- Control unit (CU)
- Accumulator
- ALU is used for arithmetic operations and for logical decisions. Memory is used for storing the instructions after processing. Control unit is used to control the internal and external peripherals which are connected to the CPU and accumulator is used for storing the results.

MEMORY ORGANIZATION

PIC microcontroller memory module consists of mainly 3 types of memories:

- PROGRAM MEMORY
- DATA MEMORY
- DATA EEPROM

MEMORY ORGANIZATION

PROGRAM MEMORY:

- It contains the written program after we burned it in microcontroller. Program
- Counter executes commands stored in the program memory, one after the other.
- Pic microcontroller can have 8K words x 14 bits of Flash program memory that can be electrically erased and reprogrammed. Whenever we burn program into the micro, we erase an old program and write a new one.

MEMORY ORGANIZATION

DATA MEMORY:

It is a RAM type which is used to store the data temporarily in its registers. The RAM memory is classified into banks. Each bank extends up to 7Fh (128 bytes). Number of banks may vary depending on the microcontroller. PIC16F84 has only two banks. Banks contain Special Function Registers (SFR) and General Purpose Registers (GPR). The lower locations of each bank are reserved for the Special Function Registers and upper locations are for General Purpose Registers

MEMORY ORGANIZATION

DATA MEMORY:

These registers are used for special purposes and they cannot be used as normal registers. Their function is set at the time of manufacturing. They perform the function assigned to them and user cannot change the function of SFR. Three important SFRs for programing are:

STATUS register: It changes the bank

PORT registers: It assigns logic values 0 or 1 to the ports

TRIS registers: It is a data direction register for input and output

MEMORY ORGANIZATION

DATA EEPROM:

This memory allows storing the variables as a result of burning the written program.

It is readable and writable during normal operation (over the full VDD range).

This memory is not directly mapped in the register file. It is indirectly addressed through the SFRs.

There are six SFRs which are used to read and write to this memory (EECON1,

EECON2,

EEDATA,

EEDATH,

EEADR,

EEADRH

SERIAL COMMUNICATION

The transfer of one bit of data at time consecutively over a communication channel is called Serial Communication. There are three protocols of serial communication:

USART: Universal synchronous and Asynchronous Receiver and Transmitter

SPI Protocol: SPI stands for Serial Peripheral Interface.

I2C Protocol: I2C stands for Inter Integrated Circuit,

SERIAL COMMUNICATION

USART:

It stands for Universal synchronous and Asynchronous Receiver and Transmitter which provides a serial communication in two devices.

In this protocol data is transmitted and received bit by bit through a single wire according to the clock pulses.

To send and receive data serially the PIC microcontroller has two pins TXD and RXD.

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SERIAL COMMUNICATION

SPI Protocol: SPI stands for Serial Peripheral Interface.

It is used to send data between PIC microcontrollers and other peripherals like sensors, shift registers and SD cards.

Three wire SPI communications is supported in PIC microcontroller between two devices on a common clock sourc

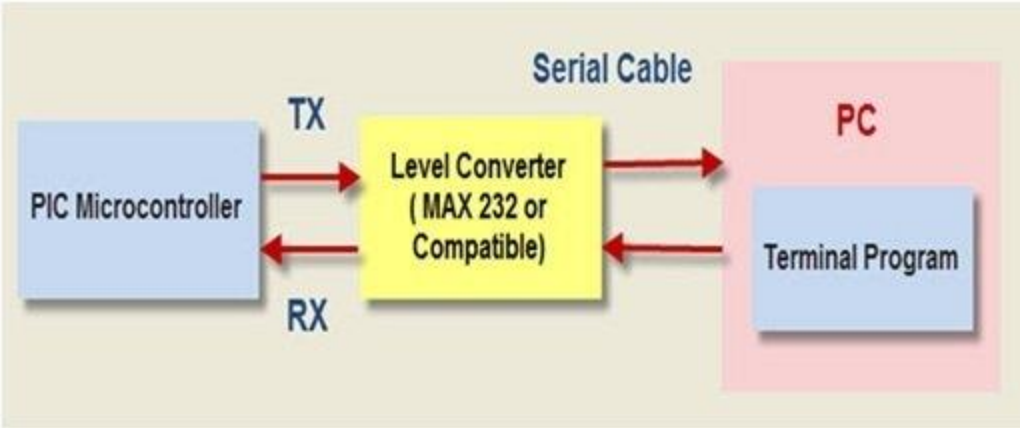
SPI protocol has greater data handling capability than that of the USART.

SERIAL COMMUNICATION

I2C Protocol: I2C stands for Inter Integrated Circuit,

this protocol is used to connect low speed devices like microcontrollers, EEPROMS and A/D converters.

PIC microcontroller support two wire Interface or I2C communication between two devices which can work as both Master and Slave device.



INTERRUPTS

There are 20 internal interrupts and

three external interrupt sources in PIC microcontrollers which are related with different peripherals like ADC, USART, Timers, and CCP etc.

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PORTS

Let us take PIC16 series, it consists of five ports, such as Port A, Port B, Port C, Port D and Port E.

Port A:This port is 7-bit wide and can be used for both input and output. The status of TRISA register decided whether it is used as input or output port.

Port B:It is an 8-bit port. This port also can be used as input and output. Moreover in input mode four of its bits are variable according to the interrupt signals.

Port C:It is also an 8-bit port and can be used as both input and output port which is determined by the status of the TRISC register.

Port D:This 8-bit port, unlike Port A, B and C is not an input/output port, but is used as acts as a slave port for the connection to the microprocessor When in I/O mode Port D all pins should have Schmitt Trigger buffers.

Port E:It is a 3-bit port which is used as the additional feature of the control signals to the A/D converter.

CCP MODULE

A CCP module works in the following three modes:

Capture Mode: In this mode time is captured when a signal is arrived, or we can say that, when the CCP pin goes high it captures the value of the Timer1.

Compare Mode: It works same as an analog comparator, which means that when timer 1's value reaches some reference value it will give an output signal.

PWM Mode: This mode provides a 10 bit resolution pulse and duty cycle that is programmable.

COUNTERS

Timers and counters are important as timers can tell the time and count.

PIC microcontroller can have up to four timers (depending upon the family)

Timer0, There are no analog outputs in PIC Microcontroller. To get analog output we have to use external Digital-to-Analog Converter (DAC). It can convert 8 bits of digital number from the eight digital outputs of PIC microcontroller.

Timer1

Timer2 and

Timer3.

Timer0 and Timer2 are of 8-bits while the

Timer1 and Timer3 are of 16-bits, which can also be used as a counter.

These timers work according to the selected modes.

DAC

There are no analog outputs in PIC Microcontroller.

To get analog output we have to use external Digital-to-Analog Converter (DAC).

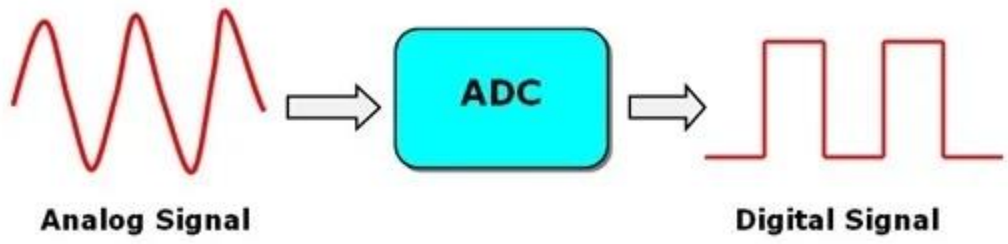
It can convert 8 bits of digital number from the eight digital outputs of PIC microcontroller.

ADC

It converts the analog voltage levels to digital voltage values.

In PIC Microcontroller, ADC has 8-channels and has resolution of 10-bit, which means that if we have to convert an analog voltage between 0V to 5V the converter will divide it to 2^10 levels (1024 levels).

The special function registers ADCON0 and ADCON1 control the operation of ADC. The converter stores the lower 8 bits in ADRESL register and the upper bits in the ADRESH register. Reference voltage of 5V is required for the operation of the converter



Features of PIC18

- **CPU:** 8 bit Microcontroller
- **Program Memory :** 2 mb
- **Data Memory:** 4 kb
- **Flash ROM:** 32 kb to 128 kb
- **EEPROM:** 256 bytes to 1 kb.
- **Operating Voltage:** 4.2 to 5.5 volts. (2.5 volt)
- **Pin Package:** 40 to 44 pins.
- **Instructions Set**: 35 smaller.
- **Internal Hardware Timer:** 4
- **Interrupt Sources:** 18
- **A/D Converter:** 10 bit
- **Max. CPU Speed (MHz):** 40 MHz

- **Performance:** 10 MIPS-16 MIPS
- **Address Bus:** 21 bit in Program Memory.

  12 bit in Data Memory.
- **Data Bus:** 16 bit in Program Memory.

  8 bit in Data Memory.

Comparison of PIC Family

PIC Family	Pins	Flash/ RAM Memory range	Timers	I/O pins/ ADC channels	Additional Features
PIC 10CXX	6	896 B/ 64B	8 bit(2), PWM (2)	4/ 2ch. 8bit	---
PIC 12CXX	8	3.5 KB/128-256 B	8 bit(2), 16 BIT(1)PWM (2)	6/ 3 ch.8 bit	USART(1)
PIC 16FXX	14-40	7-28 KB/512B-2KB	8 bit(4), 16 BIT(1)PWM (2)	12/ 4 ch 10 bit,	WDT, Data E, USART, I2C, SPIEPROM
PIC 18FXX	40-100	2 MB/32-128 KB	3 Timers, WDT, PWM	33- 72/ 12 ch 10 bit	USB, 12 bit ADC USART CAN , I2C, SPI Bus,

New for 18 series

- The number of instructions more than doubled, with 16-bit instruction word
- Enhanced Status register
- Hardware 8 × 8 multiply
- More external interrupts
- Two prioritized interrupt vectors
- Radically different approach to memory structures, with increased memory size
- Bigger Stack, with some user access and control